

Project Presentation Content (10-Slide Version)

Condensed from `project_description.md` for a 10-slide PPT. One slide = one heading below.

Slide 1 — Title

Smart Digital Complaint Management and Public Transparency System

Account-less civic issue reporting, public audit trails, and AI-assisted photo validation

- Web portal for citizens to report public infrastructure issues (potholes, garbage, water leakage, streetlights, and 6 more categories)
- No account required — email OTP verification only
- Public registry + admin console + proposed AI validation layer

Visuals:



Slide 2 — Problem Identification

- Civic issues are reported today via phone hotlines, in-person visits, or informal social posts
- No tracking ID, no public record, no category-specific urgency, no photo verification
- Result: lost/duplicate complaints, inconsistent triage, no public accountability

Problem Statement: Civic issue reporting is fragmented, untracked, and invisible to the public.

Visuals:



Slide 3 — Literature Survey & Research Gap

Prior Approach	Strength	Limitation
Phone hotlines	Centralized entry point	No photo evidence, no public tracking
Crowdsourced issue maps	Public visibility	No identity check, no urgency scoring
Generic helpdesk ticketing	Mature status workflow	Not citizen-facing / not transparent
CV-based defect detection	Automated visual check	Needs category-specific training data

Visuals:



Slide 4 — Objectives

Primary Objective: Design a transparent, account-less civic complaint platform with severity-aware triage and AI-assisted photo validation.

- Accessibility — OTP verification, no account
- Structured categorization — fixed 10-category taxonomy
- Intelligent prioritization — weighted questionnaire score
- Automated visual validation (*proposed*) — per-category AI detection

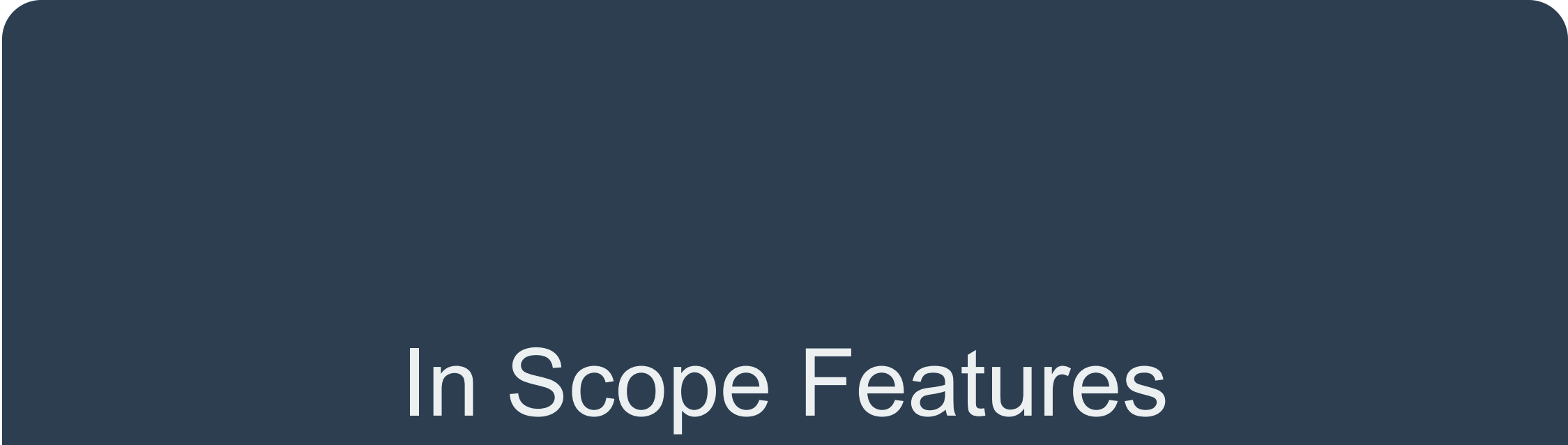
Visuals:



Slide 5 — Scope

- **In Scope:** OTP submission, 10-category questionnaire, priority scoring, public registry, admin status workflow, PDF receipts
- **Out of Scope:** native apps, chat, payments, multi-language
- **Future Scope:** production AI validation service, auto-routing, analytics

Visuals:



In Scope Features

Slide 6 — Methodology

1. Problem Identification & Requirement Analysis
2. System Design & Architecture
3. Development, Integration & Testing
4. Deployment & Demo

Visuals:

 Phase 1

Phase 2: Development

Slide 7 — System Architecture

- **Frontend:** React (Vite) — citizen wizard, public registry, admin console
- **Backend:** Express.js / Node.js — REST API, OTP, status workflow
- **Database:** MongoDB — Users, Complaints, OTPs (TTL)

Visuals:

Frontend: React + Vite

Slide 8 — Proposed AI Validation Layer

- One YOLO-based object-detection model per category (e.g. `pothole_road_damage_model.pt`)
- Detects the issue, draws a bounding box, returns confidence + annotated image
- Low-confidence detections are routed to an admin review queue

Visuals:

YOLO Detection Bounding Box

Slide 9 — User Workflow

- Select Category & Questionnaire (Swiping Yes/No)
- Details + Photos + Contact Info
- Email OTP Verification
- Tracking ID Issued -> Public Tracking

Visuals:

App UI: Swiping Yes/No

Slide 10 — Conclusion & Final Architecture

- Replaces fragmented, untracked civic reporting with one transparent, OTP-verified, auditable platform
- Weighted questionnaire delivers consistent, explainable priority triage
- Proposed AI layer adds automated visual validation

Visuals:



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